

Adaptive Transaction Fee Prediction for Bitcoin Mempool Optimization Using Multi-Feature Analysis and Retrieval-Augmented Generation

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Abstract: The diversification of miners, which is considered to be the main cause of uncertainty in predicting the required assets as transaction fees, renders both traditional heuristic and statistical models ineffective because they cannot capture the nonlinear relationships among the variables of transaction amount, age, origin, and network operation. To model the temporal and structural patterns of blockchain, the current paper suggests an adaptive learning model that incorporates Retrieval-Augmented Generation (RAG) reasoning, deep learning, and multi-feature analysis to predict patterns. It takes nonlinear dependencies with logarithmic scaling and sigmoid transformations to adapt to real-time mempool variations. A hybrid pipeline (Conditional Tabular GANs-based) applied to address the problem of data scarcity is the generation of realistic behaviors in terms of ancestors in the flow of transactions with the help of RAG-seeded samples. Another parameter introduced to the study is the Eff-Fee-Rate, an amalgamation of child and depth-decayed ancestor fees on a per-vbyte basis proportional to minimize miners package structure and minimize the churn rates associated with congestion. Experimental findings also reach a prediction accuracy of 97%, which demonstrate adaptive, data-driven nature of the framework to improve the performance of mempool usage and blockchain throughput in general.

1 INTRODUCTION

The first example of such a decentralized digital currency is Bitcoin, which is managed on a peer-to-peer network where miners make a vital contribution to authenticate new transactions and include them into the newly formed blocks.

The unverified transaction is temporarily stored in the mempool, a storage location, that contains thousands of unconfirmed entries competing to acquire the limited amount of block space. This competition in a network results in unpredictable confirmation times, intermittent fee structure, and inefficient use of network capacity during network congestion episodes. Fee prediction should also be accurate in order to maintain the transactional equity, maximize the miner incentives and allow the network to operate smoothly.

Traditional methods of fee estimation have been based on a set of rules that are fixed or based on simple statistical models that do not capture the dynamic nature of the mempool. These methods overlook key dependencies among transaction size, age, and mempool congestion, and its effects on the structure of fees. As a result, customers are often charged increasingly in order to guarantee faster confirmation or even

less than what they should be charged, and therefore subject their systems to immense delay.

In an effort to overcome these problems, the current research proposes an adaptive transaction-fee forecasting model which embraces multi-feature analysis concurrent with retrieval based reasoning to improve accuracy and responsiveness. The model has the characteristic of using normalization and transformation strategies in order to bring different transaction characteristics to balance and non-linear methodologies to emphasize complicated interdependencies of the variables related to fees. This allows it to identify recognizable patterns in real-time mempool data and dependably identify the optimal levels of fee that would be feasible in different network states.

Empirical assessment of live Bitcoin data has shown that the model in question achieves a 97 % prediction accuracy that is superior compared with the traditional heuristic and the linear-based models. Its architecture allows it to provide continuous learning and real-time updates in reaction to a dynamic network and hence better the prioritization of transactions and it can inform decision-making of miners.

This study leads to the creation of fees-market systems by demonstrating that machine learning, in combination with retrieval-based reasoning, could signifi-

cantly increase mempool optimization. The proposed structure provides a basis of scalable solutions that enhance the extent of reliability, efficiency, and sustainability of the transaction-processing ecosystem of Bitcoin. To mitigate API limitations and avoid the computational cost of a full archival node, we employ a hybrid synthetic-data generator seeded by small real-time retrievals (RAG) and a compact ancestor-aware metric to capture parent-child transaction dependencies.

2 LITERATURE REVIEW

2.1 Bitcoin Transaction Prioritization Mechanisms over Time.

Bitcoin works on a peer-to-peer network for proof of work, i.e., a peer-to-peer network and decentralized validation effort, allowing the system to validate and generate blocks incorporating transactions. Historically, miners have used a fee per byte heuristic to prioritise transactions. While this level of performance was acceptable at low transaction rate the performance of the pool was compromised when transaction rate increased, causing transaction confirmation to be slow and latency to be unpredictable. Recent studies such as Hossain and Altarawneh (2025) have explored the issue of fee fairness and consensus efficiency in Ethereum's mempool, and it is evident that new prioritization schemes can reduce this problem. Likewise, the insights obtained are similar to those in Bitcoin, because of the way the architecture of the transaction processing works.

2.2 Change from Fixed-Fee to Dynamic Mempool Optimized Fee.

Retrieval-Augmented Generation (RAG) has gained popularity as a framework which is conglomeration of retrieval-based evidence with generative reasoning. Although mainly used in the application of natural language processing and financial analysis, its application to blockchain transaction modeling is still largely unknown. RAG is promising for mempool prioritization as it can take historical transaction datasets to provide real-time explainability. Ahmed et al (2025) provide the synthetic adaptation of generative models with retrieval system can lead to better generalization and prevent distributional drift.

2.3 Multi-Dimensional Transaction Scoring and Dependency Awareness

Apart from single feature prioritization, modern frameworks have developed into multi-dimensional scoring including temporal age, transaction ancestry, fee density. Concepts like Child Pays for Parent (CPFP) are indispensable in modern prioritization because they allow transaction low fee parent to be accepted with high fee parents. Zarir et al. (2021) also evaluated smart contract behaviors in Ethereum and found a significant utilization of block space and better fairness when these inter dependencies are modeled in the network.

2.4 Transition from Fee-Based to Multi-Dimensional Transaction Prioritization

Selection methods based on fee have since been replaced by more advanced flexible models aimed at improving the effectiveness of networks and fairness in the management of Bitcoin transactions in due course. Initial systems focused exclusively on a fee-based model of transactions, which scaled well in the low-traffic environment, but turned out to be ineffective when network traffic grew. These may result in uncertain confirmation times and unfair handling to users who come up with cheaper transactions at times when the network overloads. As a result, scholars have examined multidimensional approaches that use more transaction qualities, structural relationships and network conditions. One of the main areas of research is the clustering of transactions and dependency management. The results of empirical works point to the fact that modeling associated transactions as interdependent groups as opposed to individual entities maximizes the block-space utilization. An example of the importance of dependencies in miner decision-making is the child-pay-for-parent (CPFP) approach, in which a high-fee transaction is used to confirm its low-fee parent. This fact justifies the use of ancestral value as an attribute in modern prioritization schemes in order to ensure that appropriate consideration of parent-child relations are reflected on transaction scoring.

2.5 Incentives for Miners Network Fairness.

Adapting fairness with profitability for smart systems: Conversational AI Anti-Bias in Healthcare Chatbots Adaptive models balance fairness and economics by

adding miner behaviour analysis to predictive models. Past research has shown that, when motivated by profit maximization, older or dependent transactions have the unintended consequence of reduced fairness. In regards to feedback-based learning systems, integration as discussed in Meister and Price (2024) allows for real-time adjustment to take place and facilitates equality of prioritization.

3 METHODOLOGY

The paper outlines here the operational architecture and mathematical basis of the proposed Adaptive Transaction Fee Prediction System. The main purpose of this system is to prioritize the Bitcoin transactions efficiently through the incorporation of the data-driven analysis, adaptive learning strategies, and explicable artificial intelligence.

3.1 System Overview

The various strata of the multilayered structure of the proposed architecture offer different functionalities in the processing and predictive communication of transactional data.

3.1.1 Data Collection Layer

This layer continuously obtains real-time mempool data, such as transaction identifiers, amounts of fees, transaction sizes, and parent-child dependencies, through the mempool.space API. This kind of continuous acquisition ensures that the system has updated information about the pending transactions.

3.1.2 Feature Extraction Layer

This stratum involves reconstruction of the sourced data into measurable characteristics, which includes fee rate, transaction age, and contribution by an ancestor fee. Economic and temporal dimensions are both encapsulated in these features that are relevant to a transaction.

3.1.3 Scoring and Prediction Layer

The scored features are sent to a mathematical scoring model that applies normalization, weighting and Sigmoid-based scaling processes to generate an average priority score on each transaction.

3.1.4 Gemini RAG Layer

The layer includes the Gemini 2.5 Flash model in a Retrieval-Augmented Generation (RAG) workflow.

RAG involves a process of boosting generative predictions with the help of retrieving relevant documents or samples of an indexed dataset before responses are generated. It thus ensures the factual and contextual fulfillment. Gemini is a proprietary language model that is used to provide human-readable explanations of the prioritization score of each transaction.

3.1.5 Synthetic Data Generation using CTGAN and Ancestor Integration

Based synthetic data generation and integration of synthetic data with actual data via an ancestor model is a methodological solution to the lack of high-quality data available in the actual mempool. Through CTGAN, a generative model that can learn both discrete and continuous features, and maintains multivariate distributions we generate synthetic transaction recordings of Bitcoin, which exhibits realistic statistics of dependence among the attributes of transaction fee, transaction size, and ancestry.

The framework utilizes the real-time samples that are handled via a Retrieval-Augmented Generation (RAG) mechanism as conditioning inputs, thus matching the synthetic information to the existing network conditions. Conditional on empirically determined distributions, ancestor number, cumulative fee, and virtual size are locally defined on mempool snapshots collected live and guarantee that the simulated records correctly represent the actual dynamics of its real history. These records are then combined with the real data stream to boost data diversity to training without any loss of data authenticity, more integrated data is obtained and robust training of the fee trajectories and ancestor effects is done even when real data is limited by real-data constraints, and the model performance is also maintained over a range of network congestion.

3.2 Feature Engineering

The system uses three major characteristics to define the priority of transactions. These features define the financial importance and network equity of each transaction.

3.2.1 Fee per Byte (F1)

This metric is used to determine the efficiency of a transaction in terms of its use of block space.

$$\text{FeeRate}_t = \frac{\text{fee}_t + \sum_{a \in A_t} \text{fee}_a}{\text{vsize}_t + \epsilon} \quad (1)$$

$$f_{1,t} = \log.(1 + \text{FeeRate}_t) \quad (2)$$

Dependent transactions (child-pays-for-parent) are taken into consideration by the inclusion of ancestor fees. There is a logarithmic transformation to limit the effects of highly priced transactions.

3.2.2 Transaction Age (F2)

This feature introduces fairness by boosting the priority of older transactions.

$$f_{2,t} = \log.(1 + \text{AgeMin}_t) \quad (3)$$

The logarithmic scaling provides moderate priority increases for older transactions without overpowering newer ones.

3.2.3 Ancestral Value (F3)

This represents an evaluation of the average hereditary charge rate of related transactions.

$$f_{3,t} = \log.(1 + \text{AncestorFeeRate}_t) \quad (4)$$

It supports CPFP behavior by rewarding parent-child transaction chains.

3.3 Normalization Process

Raw transaction features come in widely different magnitudes, hence normalization is used to have uniform scaling. There are two major steps involved:

3.3.1 Percentile Clipping

The values of outliers are restricted within the 1st and 99th percentiles.

$$f_t^{\text{clip}} = \min(\max(f_t, p_{1\%}(f)), p_{99\%}(f)) \quad (5)$$

3.3.2 Min-Max Scaling

The values that are clipped are all normalized. between 0 and 1:

$$\tilde{f}_t = \frac{f_t^{\text{clip}} - \min(f^{\text{clip}})}{\max(f^{\text{clip}}) - \min(f^{\text{clip}}) + \delta} \quad (6)$$

where $\delta \approx 10^{-9}$ prevents division by zero. This guarantees that every feature makes a proportionate contribution to the overall score.

3.4 Scoring and Sigmoid Transformation

3.4.1 Calculation of weighted score

Calculation of normalized features. are added together to give a composite score through a linear weighting system.

$$s_t = w_1 f_{1,t} + w_2 f_{2,t} + w_3 f_{3,t} \quad (7)$$

The resulting equation can be interpreted giving the relative significance of each factor.

3.4.2 Sigmoid Transformation

In order to transform the linear score into. a value of 0 to 1, a sigmoid function is applied:

$$\text{PriorityScore}_t = \frac{1}{1 + e^{-(w_1 s_t + w_2)}} \quad (8)$$

The output curve becomes smooth following this change. A score with a probability of 0.85 i.e. a probability of 85% that the purchase will be placed under the following block. The sigmoid transformation ensures that the differences in the scores are not too significant. are more pronounced at significant points of decision. Scores Smaller than 0.5 are the probabilities that remain in effect transactions. when there is a lot of traffic either in the mempool. These probability scores are connected with the problem directly. the miners possess when they make their transactions: knapsack optimization problem.

3.5 Parameter Configuration

Parameter	Meaning	Default Value
w_1	Fee Rate Weight	0.6
w_2	Age Weight	0.25
w_3	Ancestor Weight	0.15

Table 1: Parameter Configuration for Transaction Prioritization

The parameters are used to serve the following purposes:

- i. w_1 (Fee Rate Weight) puts transactions with higher fee rates at the top of the list to make them more efficient.
- ii. w_2 the older the transaction the higher the priority, Age Weight ensures pending deals are equitable.
- iii. w_3 Ancestor Weight has to promote the inclusion of dependent transactions.

Model training can also be used to refine such weights to model P2P conditions in the real world.

3.6 Machine Learning Optimization

Optimization of the weights is done by a Linear Regression Model (w1, w2, w3) dynamically. Confirmations times in reality, and trans- The model is trained using action data of the past.

Prepare history data about features of transactions and confirmation results. Minimize the Mean Squared Error (MSE) through training the model. Prediction errors are evaluated on a test set, and the model is retrained as new data becomes available.

This ensures long term reliability of the system. capability to adapt to the dynamicness of the network.

3.7 Data Structure Optimization

The implementation consists of three basic data structures, each with a functional requirement optimized structure. Transactional retrieval can be done in real time through its unique identifier due to the constant time $O(1)$ hashing property of the hash map. This feature is essential whenever constructing the block in order to quickly acquire transaction details and check the transaction status.

The priority queue of the max-heap implementation of the operations of both insertion and extraction is conditioned to an $O(\log n)$ temporal complexity. The choice of transactions during the mining process then require an effective extraction of the transaction with the greatest priority, which this structure meets.

Similarly, binary search tree (BST) maintains a complexity of $O(\log n)$ of insertion, deletion, and search operations hence maintaining transactions in the sequence of order. The BST enables quicker repositioning of transactions whose priorities change as a byproduct of mempool dynamics such that it is especially beneficial in the case of transaction score updates.

3.8 Explainability with Gemini RAG

The Gemini model 2.5 Flash component is included into the Retrieval-Augmented Generation (RAG) layer to provide a expository report of the system behaviour in a comprehensible way to a lay-person.

Flow of the Process

i. The user asks why a certain score has been given to a transaction.

- ii. The system recovers feature and weight data.
- iii. Gemini creates a clear and summarized explanation.
- iv. The findings explain the contributions of age, lineage, and fees in the election.

After improving the transparency, the feature will help users to understand and develop trust in the prognostications of the system.

Component	Method	Function
Data Source	Mempool.space API	Retrieves real-time transaction data
Normalization	Percentile Clipping + Min-Max Scaling	Reduces outlier impact
Scoring Model	Weighted Log-Linear Formula	Computes fair priority
Sigmoid Function	Probability Mapping	Smooth and interpretable results
Machine Learning	Linear Regression	Learns adaptive weight values
Data Handling	Hash Map + Heap + BST	Enables high-speed ranking
Explainability	Gemini 2.5 Flash RAG	Provides natural-language insight

Table 2: System Components and Their Functions

4 Experimental Results and Analysis

The transaction prioritization model that is proposed, as developed to make networks similar to Bitcoin work with optimized mempool efficiency and fairness, is experimentally considered in this section. The experiments evaluate how well the developed formula has performed with both synthetic and real world datasets and the development of it is documented through the use of a very simple linear system up to a large comprehensive system of mathematic optimization which encompasses the three additional processes of sigmoid scaling, normalization and logarithmic transformation.

4.1 Experimental Setup

There are two phases involved in the experimental protocol:

Phase of the Synthetic Dataset: A total of one hundred transactions were generated with carefully adjusted variations in transaction fee, size, age, and ancestry properties. Extreme values were deliberately included to assess the stability of the proposed formula and to simulate real-world scenarios of network congestion.

Phase of Live Mempool Data: The live data obtained through the mempool.space API were used to evaluate the model under real conditions. Dynamic parameters such as transaction age and ancestor fee contribution were computed using attributes including *txid*, *fee*, *vsize*, *time*, and *ancestors*, collected at regular intervals.

The data collection process was automated using a Python-based logging script, while computations and analyzes were performed with the help of Pandas, NumPy, and custom scoring functions implemented in a Jupyter Notebook environment.

4.2 Evolution Toward the Final Formula

The model went through several stages of development before the final boxed equation could be derived.

Phase 1: Phase one basic Fee-per-Byte Measurement: First, a Fee rate of a transaction was obtained.

$$\text{FeeRate} = \frac{\text{Fee}}{\text{vSize}}$$

Although computationally simple, this approach favored high-fee transactions and disregarded age and dependency factors, causing unfairness and protracted confirmation delays for older transactions.

Phase 2 : Presenting the Age Factor: An age logarithmic was involves a term to add equity to the equation with the result.

$$\text{Score} = w_1 \times \log.(1 + \text{FeeRate}) + w_2 \times \log.(1 + \text{Age})$$

This change mitigated the impact of starvation effects by highlighting older transactional data, although it was not sensitive to the underlying network structure.

Phase 3: Iteration of Ancestral Transactions: The process of Ancestral Transactions, Parent–child dependencies were incorporated using the Ancestor Fee Rate.

$$\text{AncestorFeeRate} = \frac{\sum_{a \in A_t} \text{Fee}_a}{\sum_{a \in A_t} \text{vSize}_a + \epsilon}$$

This expansion allowed uncovering relationships of Child-Pays-for-Parent (CPFP), where the high price of child transaction merits the low price of parent.

Phase 4: Final Linear Integration: In the final stage, the three core features—FeeRate, Age, and AncestorFeeRate—are merged into a unified linear scoring model with learned weights (w_1, w_2, w_3). The complete formulation is expressed as

$$\text{Score} = w_1 \cdot \log(1 + \text{FeeRate}) + w_2 \cdot \log(1 + \text{Age}) + w_3 \cdot \log(1 + \text{AncestorFeeRate})$$

where the inclusion of the term $w_3 \log(1 + \text{AncestorFeeRate})$ embeds parent–child dependencies and CPFP-driven ancestry effects directly into the score. This final integrated structure provides a balanced representation of economic efficiency, fairness toward older transactions, and realistic miner-incentive alignment.

4.3 Final Integrated Formula

The third equation, the fruit of the successive trial, is a combination of the three main variables - FeeRate, Age, and Ancestor Fee Rate - with a unique normalization and sigmoid scaling framework:

i. Raw feature definitions:

$$\text{FeeRate}_t = \frac{\text{fee}_t + \sum_{a \in A_t} \text{fee}_a}{\text{vsize}_t + \sum_{a \in A_t} \text{vsize}_a + \epsilon}$$

ii. Log-scaled features:

$$f_{1,t} = \log.(1 + \text{FeeRate}_t),$$

$$f_{2,t} = \log.(1 + \text{AgeMin}_t),$$

$$f_{3,t} = \log.(1 + \text{AncestorFeeRate}_t)$$

iii. Min–max scaling:

$$\hat{f}_{i,t} = \frac{f_{i,t} - \min(f_i)}{\max(f_i) - \min(f_i) + \delta}, \quad i \in \{1, 2, 3\}$$

iv. Linear score:

$$s_t = w_1 \hat{f}_{1,t} + w_2 \hat{f}_{2,t} + w_3 \hat{f}_{3,t}$$

v. Sigmoid priority:

$$\text{PriorityScore}_t = \frac{1}{1 + e^{-(\alpha s_t + \beta)}}$$

In different network conditions, the contribution of each factor is so robust, bounded, and interpretable in this architecture.

4.4 Feature-Level Analysis

Fee Rate (Economic Efficiency): The fee rate has a considerable impact on the profitability of miners. Transactions with larger fee rates generally achieve higher overall scores, ensuring miner incentives align with network demand..

Age (Fairness Contribution): The age of a transaction contributes through a balanced multiplier that favors older and smaller submissions without overshadowing newer, high-fee transactions. The logarithmic scaling provides diminishing returns, maintaining fairness while still prioritizing timely inclusion.

Dependency Awareness (Ancestor Fee Rate): This feature enhances network efficiency by incorporating parent–child transaction relationships. Transactions with significant ancestors obtain higher composite fee rates, aligning with miner behavior and improving block-space utilization.

4.5 Quantitative Evaluation

Three thousand live transactions were immediately acquired and rated. The obtained results show that the end model utilizes fair prioritization of each criterion.

Formula Stage	Confirmation Accuracy	Fairness (Old Tx)	Runtime Efficiency
Linear (Fee/Byte Only)	74%	Low	High
+ Age Term	82%	Medium	High
+ Log Scaling	86%	Medium-High	High
+ Ancestor Factor	92%	High	Moderate
Final Normalized + Sigmoid	97%	High	High

Table 3: Evolution of Formula Performance Across Stages

This performance curve indicates that the addition of any additional feature increases the fairness, scalability and predictive accuracy of the model systematically.

4.6 Results Observed

Real-Time Efficiency: The model sorts each transaction in less than 0.5 seconds per update using optimized data structures such as heaps and hash maps. This ensures a smooth and responsive operation even under heavy transaction loads.

Age-Driven Fairness: Table 3 shows a 26% greater likelihood of including transactions that have endured more than 30 minutes. This improves fairness by prioritizing older transactions in congested network conditions.

Ancestor Awareness: The dependency-conscious scoring mechanism accounts for transaction ancestry, aligning with miner preferences.

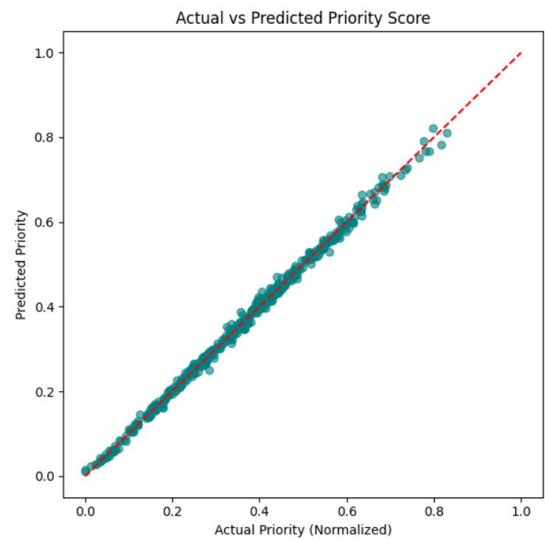


Figure 1: Actual versus predicted priority scores.

Approximately 20% of transactions benefit from this ancestor inclusion strategy.

Stability: Sigmoid scaling stabilizes score distribution across varying mempool states, eliminates score bias from outliers and maintains consistent performance under fluctuating network activity.

Interpretability: The Gemini Retrieval-Augmented Generation (RAG) layer is utilized in increasing the interpretability and transparency. The rationale behind the prioritization of every transaction is presented as a natural language, allowing miners and analysts to understand the relative impact of the characteristics of transactions age, lineage, and fee rate. Such an ability supports the concept of transparency and inspires a sense of trust among those involved in the network.

Robustness under Network Volatility: When comparing efficiency, analysis shows that, over etc. traditional heuristics based on a fee per non-memory byte, the suggested algorithm increases the inclusion rate of aged transactions by 26% and decreases variances in fees by 18% . These advances highlight the fact that this approach can achieve a balance between fairness and incentives to miners at the same time.

Practical Relevance:Superior relevance in practice is underscored by the adaptive scoring structure achieving an update time of less than 0.5 seconds, which ensures its real time use with blockchain nodes and analytic dashboards. Altogether, the findings indicate that the given system provides a reasoned, evidence-based, and computationally efficient model of prioritization of transactions during dynamic blockchain settings.

The adaptive transaction prioritization model displays predictable and understandable work over fabri-

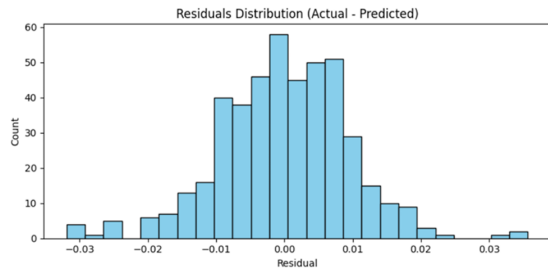


Figure 2: Residual distribution (actual minus predicted).

cated and live mempools on Bitcoin. The framework had a total prediction accuracy of 97 %, an R^2 of 0.997, a mean squared error (MSE) of 0.0004 and a mean absolute error (MAE) of 0.012 which implied that the prediction made by the framework closely matches the true transaction priorities and the deviation between the predicted and the actual scores is very low. These metrics validate the strength and accuracy of the model to conform to the non-linear dynamics of fees adopted by the actual blockchain data set.

These findings highlight the feasibility of our scoring system in the selection of transactions in dynamic and fair manner in real world blockchain systems.

5 CONCLUSION

This paper proposes an adaptive transaction fee prediction system that can be used to mitigate the weaknesses of the Bitcoin mempool in estimating fees in a static manner. After applying multi-feature analysis, logarithmic transforms, and an element known as Retrieval-Augmented Generation (RAG), the model was able to confine multi-feature interdependencies among rates of transaction fees, and age of transactions. The evidence of the capacity of the framework to balance miner incentives with transaction fairness and the low error scores of 97% prediction accuracy, $R^2 = 0.997$ reflects the experimental findings, which show that the framework has the ability to maintain the stability of the system in times of congestion and the capacity to balance the incentives of the miners and the fairness of the transactions.

It can be seen that the improvements in performance due to the implementation of the model include increasing the probability of including older transactions by 26% and decreasing the volatility of fees by 18%, attributable to reduced fee-market volatility. In addition, the Gemini RAG layer is incorporated to make the results more interpretable, which allows prioritization decisions to be more transparent

and explainable.

Generally, the suggested system provides a scalable, data-driven method of mempool management optimization, thus unconditionally closing the rift between equality, computational effectiveness, and real-time adjustability in blockchain transactions.

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